

Conflict and Peace

Week 10: Resistance - Overcoming collective action problems

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

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What is collective action?

Definition of collective action

- Action taken by a group in pursuit of perceived shared interests
 - ▶ Self-organizing protests to advocate certain causes and interests
 - ▶ Fighting in the military during wartimes
 - ▶ Negotiations / strikes carried out by labor unions (or any interest groups)
 - ▶ Management of communal resources
- Involves sharing information and coordination among individuals to pool resources
- Requires motivation and incentives from many individuals sharing common goals

This is a conceptual framework often used across many disciplines in social sciences

- Politics: How do interest groups sway votes? How do alliances sustain themselves?
- Economics: How are public goods managed? How do groups organize?

Why is it difficult to achieve?

Individual and collective incentives are misaligned

- Larger groups: Larger misalignment and smaller return per capita
 - ▶ Coordination challenges: Larger coordination costs and information costs
- Free-riding: Attempt to benefit from public good without individual contribution
 - ▶ Game theory: If everyone thinks the same, public good is underprovided
 - ▶ Tragedy of commons: Rational individuals end up depleting shared resource

So how to overcome this problem?

- Decrease coordination costs: Communications and dispute settlement mechanisms
- Align individual and collective incentives
 - ▶ Make deviations costly: Monitoring, enforcement of contracts / laws
 - ▶ Internalize collective benefits: Increase awareness and salience of collective goals
- Institutional setup to aggregate collective interests and costs effectively (governance)

However, collective actions do happen worldwide



(a) Political protests



(b) Environmental protests



(c) Community service



(d) Paying taxes

Access to media and collective action problems

Academic interest in media increasing

- Dissemination of information and mechanism for improving accountability
 - ▶ Voters can monitor politicians (i.e. align interests of principals and agents)
 - ▶ Improved transparency → make leaders behave well and foster trust in institutions
 - ▶ Better information as basis for electoral outcomes
- Social media, can help people mobilize
 - ▶ Inform users of the reality more accurately (information channel)
 - ▶ Used to coordinate and mobilize group activities (coordination channel)

Just as these positives, media can work in reverse: Thus, an active arena of research

- Potentially used as tools of repression (if government control is strong)
- Unchecked dissemination of misinformation
- Decaying trust in message may erode media's potential to lower coordination costs

Access to media and collective action problems

The Promises and Pitfalls of the Social Media–Fueled Gen-Z Protests Across Asia

Online activism strengthens movements, but it also exposes activists to structural and tactical vulnerabilities.



By Janjira Sombatpoonsiri

Published on September 30, 2025

Demonstrations spurred by a new generation of Gen-Z activists have roiled the streets of several Asian countries in recent months. Fueled by widening wealth gaps and alleged abuse of power, activists have organized via social media, demanding that their governments address deepening social and economic inequalities.

In Indonesia, protests that erupted in late August were spearheaded by gig workers employed by online delivery and transport platforms in response to welfare cuts proposed by President Prabowo Subianto. Amid rising concerns over the concentration of power in the hands of the armed forces and business oligarchs, public outrage intensified when the government gave members of Parliament housing allowances ten times greater than the minimum wage. In Nepal, similar frustration with entrenched elites drove Gen-Z demonstrators onto the streets. Using the hashtag #nenobabies, they

Janjira Sombatpoonsiri

Research Fellow, German Institute for Global and Area Studies



Roadmap for today

1. Overview of the topic

2. Understanding collective action - costs, leaders, and engagement

2.1 Ostrom (2000): Collective Action and the Evolution of Social Norms

2.2 Dippel, Heblich (2021): Leadership in Social Movements: Evidence from the "Forty-Eighters" in Civil War

2.3 Bursztyin, Cantoni, Yang, Yuchtman, Zhang (2021): Persistent Political Engagement: Social Interactions and the Dynamics of Protest Movements [student presentation]

3. Role of media in resistance and collective action

3.1 Manacorda, Tesei (2020): Liberation Technology: Mobile Phones and Political Mobilization in Africa

3.2 Guriev, Melnikov, Zhuravskaya (2021): 3G Internet and Confidence in Government

3.3 Enikolopov, Marakin, Petrova (2020): Social Media and Protest Participation: Evidence from Russia

3.4 Gagliarducci, Onorato, Sobbrío, Tabellini (2020): War of the Waves: Radio and Resistance During World War II [student presentation]

Ostrom (2000): Collective Action and the Evolution of Social Norms

Q: What explains how collective action takes place

Classic theories on when collective action happens

- Traditional thought: Groups take collective action whenever members jointly benefited
- Mancur Olson (1965): Zero contribution thesis

"Unless the number of individuals in a group is quite small, or unless there is coercion or some other special device to make individuals act in their common interest, rational, self-interested individuals will not act to achieve their common interests"
- In fairness, this explains some dilemma that we see in group action
 - ▶ Why some people free-ride on collective benefits
 - ▶ Why small interest group yield more influence than general public
- However, there are many incidences where zero contribution thesis is contradicted
 - ▶ Individuals voluntarily organized to provide mutual protection and monitor each other
- So what are the key factors that affect the likelihood of successful collective action?

What are the existing lab evidence: Rational egoist perspective

Behavior of self-interested rational individuals (Rational egoist) in public goods game

- Individuals split their income into public good contribution and private consumption
- There are large number of individuals N in this game
- Public goods yield some benefits for the individuals (i.e. individuals enjoy some share)
- But it reduces private consumption

Rational egoist predicts that equilibrium contribution is 0 (in most cases)

- Given others' contribution, individuals access benefits without paying costs
- Since every individual thinks the same way, no one contributes
- Difficult to deviate from this equilibrium
 - ▶ Suppose A contributes nonzero amount when everyone else contribute 0
 - ▶ Unless the returns to individual from public good $>$ contribution, A is worse off
 - ▶ And if the returns from public good $>$ contribution, equilibrium is not zero

What are the existing lab evidence: What is actually found

Findings from multiple lab experiments differ from rational egoist framework

- Individuals contribute 40-60% of endowments in finite games
- Contributions decay, but well above zero except for the final finite stage game
- Those who believe others will cooperate contribute to public goods themselves
- Individuals do learn to cooperate
- Face-to-Face communication increases cooperation substantially
- Individuals are willing to monitor/punish those who make below average Contributions
- Contextual framing matters (e.g. specifying sanctions, communications etc)

These findings suggest that there may be more type of people than mere rational egoists

Rebuilding collective action theory: Those who use norms

Maybe there are some 'conditional cooperators'

- Those willing to cooperate when they estimate sufficient share of others also do
- Explains large contributions at initial stage (and can motivate rational egoists to join)
- Vary in tolerance for free-riding - the most tolerant ones contribute to the end

Or there are also 'willing punishers'

- Those willing to rebuke or impose costs on non-cooperators
- Can reward active cooperators too

Emergence of these types?

- Evolutionary process: Norms on collective action $>$ individual returns for survival
- Thus, preferences adopt: People internalize cooperation as well as individual returns

Common goods as collective action problem

Common goods: How to cooperate when it is individually beneficial to use up resources

- Nonexcludable: Difficult to prevent beneficiaries from enjoying benefits
- Rivalrous: Individual consumption subtracts from benefits available to others
- Think of a fish stock at international waters
- *Note: Public goods are nonexcludable and **nonrivalrous***

Management of these situations suggests rise of cooperation in collective action problems

- Leader who oversees joint outcomes as key stimulus
- Clear boundary, well-tailored local rules
- Open to participation and communication in redistribution
- Existence of accountable monitors and graduated sanctions help sustain collective action
- Resolving conflicts through accessible low-cost, local arenas for discussion
- Multilayer governance and minimal recognition of right to organize (not unanimity)

Threats to sustained collective action

What makes sustained collective action vulnerable?

- Migration changes composition of those who cooperate and understand social norms
 - ▶ If in-migrants integrate well, these problems can be mitigated (two-way effort involved)
- Lack of large-scale institutional and governance arrangements to settle disputes
- Increasing opportunistic behavior through corruption
- Too much reliance on external sources that are unaware of local contexts
- Failure in horizontal/intergenerational transmission of norms through communication
- National governments imposing single set of rules on all units of governance
- Not adapting well to technological change

Growing need to understand how contextual factors affect cooperation and sustain collective action

Dippel, Heblich (2021):
Leadership in Social Movements: Evidence from
the "Forty-Eighters" in Civil War

Q: How do leaders contribute to social movements?

How do you enlist people to join a costly endeavor?

- Economic motives might be weak: Low and irregular pay
- Legal monitoring may sometimes be absent (especially wartime)
- Yet, we see many contemporary and historical examples of joining a costly endeavor
- So what brings these people to this cause?

This paper uses introduction of Forty-Eighters to study effects of leaders on enlistments

- German revolutionaries who supported egalitarian causes expelled for their activities
- Began campaigning for anti-slavery after 1854 (Kansas-Nebraska Act)
- Huge campaigners against slavery and mobilizers of Union Army volunteers
- Influence participation through local newspapers and social clubs

How people fought in the US Civil War 1861-1865

US civil war

- Fought over slavery and Southern secession
- 1/5 Adult men involved in fighting, 620,000 died in battle

So how did Germans get involved?

- German revolution (1848-1849): Many protested for republican and egalitarian causes
- Systematically prosecuted and expelled to US - hence called 'Forty-Eighters'
- Many Forty-Eighters argued for emancipation following Kansas-Nebraska Act (1854)
 - ▶ Slavery adopted based on popular sovereignty (raising salience of slavery)
- Circulated arguments and swayed votes against slavery, later involved in enlistments
 - ▶ Newspapers, Social clubs, public speeches, enlisted themselves to Union Army

Data and empirical strategy

Data sources (town-level data)

- Forty-Eighters: 493 Individual Biographies and genealogical source
 - ▶ Identify locations of their residence by 1856
- Town characteristics from 1850, 1860 Census extracts
- Union Army enlistment: Digitalized Union Army Registers

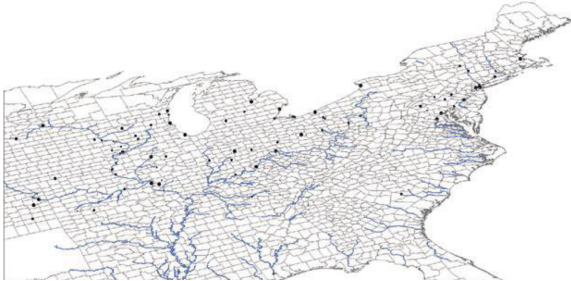
Compare Forty-Eighters towns matched with similar non-Forty-Eighters towns

- Match control group towns that share similar distribution of covariates
 - ▶ This is called propensity score matching
- Compares the matched towns with this equation

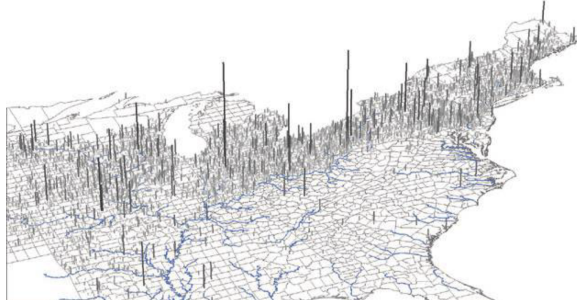
$$Y_i = \beta D[\text{FortyEighter}_i] + \gamma X_i + \phi_s + \epsilon_i$$

- Also complemented with IV: Quasi-random destination distribution of Forty-Eighters
 - ▶ Not enough time to select own destination, so followed co-passengers' destination

Enlistment and Forty-Eighters settlements match closely



(a) Forty-Eighters settlements



(b) Enlistments in Union Army

Enlistment and Forty-Eighters settlements match closely

	Full sample			Prop. score matched (PSM-1)		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>D(Forty-Eighters)</i>	0.915 [0.000] {0.000}	0.912 [0.000] {0.000}	0.894 [0.000] {0.000}	0.583 [0.001] {0.006}	0.605 [0.000] {0.004}	0.597 [0.000] {0.006}
Observations	11,095	11,095	11,095	261	261	261
R^2	0.563	0.566	0.571	0.700	0.724	0.733
Number of Forty-Eighter towns	72	72	72	54	54	54
Mean outcome	3.898	3.898	3.898	4.810	4.810	4.810
Oster delta	0.517	0.518	0.516	1.707	1.962	1.955
Core controls	✓	✓	✓	✓	✓	✓
+ other Vselect		✓	✓		✓	✓
+ all controls			✓			✓

Newspaper circulation and social clubs explain the effects

	<i>D</i> (German newsp.)		<i>D</i> (Turner Society)		log enlistments			
<i>p</i> -values in the [] brackets	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>D</i> (Forty-Eighters)	0.222 [0.003]	0.131 [0.006]	0.127 [0.018]	0.139 [0.004]	0.605 [0.000]	0.484 [0.000]	0.551 [0.000]	0.450 [0.000]
<i>D</i> (German newspapers)						0.544 [0.003]		0.491 [0.007]
<i>D</i> (Turner Society)							0.428 [0.009]	0.362 [0.017]
Fixed effects	State	Town + State $\times$ <i>t</i>	State	Town + State $\times$ <i>t</i>	State	State	State	State
Observations	261	5,376	261	5,376	261	261	261	261
<i>R</i> ²	0.576	0.739	0.333	0.477	0.724	0.730	0.728	0.733

Takeaways and remaining questions

Other results

- IV results are consistent with previous results
- Forty-eighters enlisted in Union Army themselves: Desertions ↓ in those companies
- Long-run: More likely to become members of groups advocating civil rights (NAACP)

Discussion

- Grassroots leadership can spread social movements
- They serve as focal points in motivating belief through information
- Emphasis on role of leaders are actively studied in Economics
 - ▶ How do politicians/leaders affect performances of their groups?
 - ▶ Firm profits, economic growth, institutional stability etc
 - ▶ Possible to use episodes of sudden absence to show how leaders matter

Bursztyin, Cantoni, Yang, Yuchtman, and Zhang (2021):
Persistent Political Engagement: Social Interactions and
the Dynamics of Protest Movements [student
presentation]

Manacorda, Tesei (2020):
Liberation Technology: Mobile Phones and
Political Mobilization in Africa

Q: Can digital information foster political mobilization

Are mobile technologies 'liberation technologies'?

- Can they assist marginalized groups in overcoming collective action problems?
 - ▶ Low-cost, decentralized, multi-way communication can foster activism and mobilization
 - ▶ Especially when formal political participation is blocked or repressed
- However, this may not always increase mobilization
 - ▶ It may strengthen government surveillance and repression
 - ▶ May increase government accountability through information and greater transparency

This paper analyzes whether mobile networks foster political mobilization in Africa

- Rapid mobile network expansion in 1998-2012 despite limited landline infrastructure
- Rise of mobilizations motivated by grievances (foot riots, civil unrest like Arab spring)
- Utilize georeferenced data on mobile coverage and uprisings

Development of networks and protests in Africa

How mobile phones helped protests in Africa

- Arab Spring spread by video of self-immolation of Tunisian vendor
 - ▶ The vendor protested authorities' seizure of his goods after refusing to pay bride
 - ▶ Circulated online and eventually to Al Jazeera
 - ▶ Protests erupted, fueled by discontent with low living standards, corruptions
- This spilled over to Egypt, where mobile penetration was highest in Africa
- Egyption President Mubarak resigned, despite efforts to shutdown internet

Other evidence suggest that mobile phones channeled grievances to protests

- Food riots in 2007-2012 across 14 countries spread with mobile messages and video recordings of police repression
- Mobile coverage on average was 60% higher in countries with food riots

Development of networks and protests in Africa

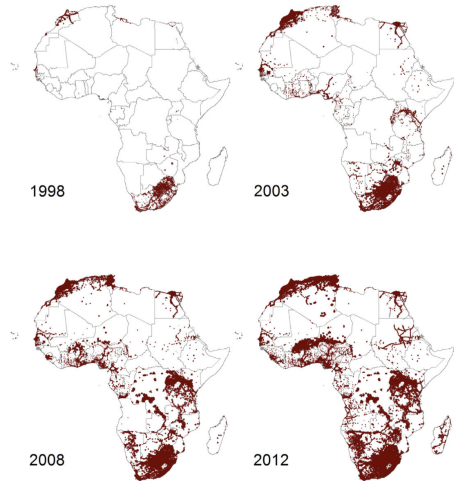
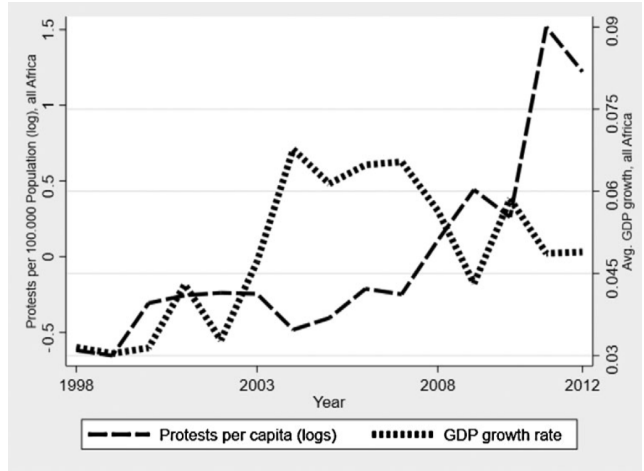


FIGURE 1.—Mobile phone coverage diffusion in Africa: 1998–2012. The figure reports georeferenced data on mobile phone coverage for all of Africa at 5-year intervals between 1998 and 2012. Source: GSMA.

(a) Mobile network coverage



(b) Continentwide Protests and GDP

Data and empirical strategy

Data sources (Georeferenced unit: 55 km $\times$ 55 km grid)

- Mobile coverage: GSMA dataset of 2G, 3G, 4G coverage (1998-2012)
- Protests $\rightarrow$ GDELT, ACLED, SCAD, Participation $\rightarrow$ Afrobarometer surveys
- Control for population, nightlights, rainfall, temperature, natural resources etc

Leverage geographical (cell j in country c) and yearly (t) variation in coverage and protests

$$y_{jct} = \beta_1 \text{Cov}_{jct} + \beta_2 \text{Cov}_{jct} \times \Delta \text{GDP}_{ct} + \gamma X_{jct} + f_j + f_{ct} + u_{jct}$$

- However, mobile coverage may not be random - instrument Cov_{jct} with lightning strikes
 - ▶ Frequent lightning damages mobile networks(negatively correlated instrument)
- Thus the first stages are

$$\begin{aligned}\text{Cov}_{jct} &= \delta_0 + \delta_1 \text{ln} n_{jc} \times t + \delta_2 \text{ln} n_{jct} \times \Delta \text{GDP}_{ct} + \gamma_1 X_{jct} + f_j + f_{ct} + e_{jct} \\ \Delta \text{GDP}_{ct} \text{Cov}_{jct} &= \theta_0 + \theta_1 \text{ln} n_{jc} \times t + \theta_2 \text{ln} n_{jct} \times \Delta \text{GDP}_{ct} + \gamma_2 X_{jct} + f_j + f_{ct} + \epsilon_{jct}\end{aligned}$$

Protests in covered areas respond to economic shocks

MOBILE PHONES AND PROTESTS: 2SLS^a

p-values reported in
brackets

	First Stage		2SLS					
	Coverage (1)	$\Delta\text{GDP} \times \text{Coverage}$ (2)	Protests (3)	GDELT (4)	Protests (5)	ACLED (6)	Protests (7)	SCAD (8)
Z	-0.006 [0.05]	0.001 [0.12]						
$\Delta\text{GDP} \times Z$	0.008 [0.67]	-0.015 [0.08]						
Coverage			-0.876 [0.15]	-0.493 [0.18]	-0.079 [0.80]	0.063 [0.85]	-0.187 [0.26]	-0.062 [0.63]
$\Delta\text{GDP} \times \text{coverage}$				-5.776 [0.03]		-2.151 [0.02]		-1.886 [0.02]
$\text{SW } F - Z$		11.580						
$\text{SW } F - \Delta\text{GDP} \times Z$		6.968						
Endogeneity test p -value			0.24	0.18	0.77	0.16	0.33	0.02
Observations	150,883	150,883	150,883	150,883	150,883	150,883	150,883	150,883

Heterogeneities across different contexts

HETEROGENEOUS EFFECTS OF MOBILE PHONES AND PROTESTS: 2SLS^a

	Institutions		Media		Pop. Density		Past Conflict	
	Democr. (1)	Autocr. (2)	Free (3)	Captured (4)	Low (5)	High (6)	No Violence (7)	Violence (8)
<u>GDELT</u>								
Coverage	-0.543 [0.78]	-1.006 [0.57]	1.118 [0.20]	-0.455 [0.17]	-2.186 [0.14]	-1.599 [0.30]	1.285 [0.07]	-2.795 [0.24]
$\Delta\text{GDP} \times \text{coverage}$	-3.055 [0.72]	-7.007 [0.16]	2.359 [0.82]	-5.973 [0.03]	3.878 [0.56]	-10.299 [0.11]	-3.466 [0.27]	-2.367 [0.10]
<u>ACLED</u>								
Coverage	2.577 [0.49]	-0.770 [0.61]	1.219 [0.08]	-0.231 [0.46]	-0.374 [0.43]	0.001 [0.99]	0.278 [0.25]	-0.010 [0.99]
$\Delta\text{GDP} \times \text{coverage}$	0.761 [0.94]	-2.232 [0.35]	-2.930 [0.39]	-1.790 [0.10]	3.554 [0.38]	-3.076 [0.06]	-0.987 [0.20]	-3.040 [0.04]
<u>SCAD</u>								
Coverage	0.706 [0.36]	0.233 [0.49]	0.354 [0.31]	-0.063 [0.60]	-0.600 [0.18]	0.073 [0.77]	-0.068 [0.56]	-0.241 [0.18]
$\Delta\text{GDP} \times \text{coverage}$	-0.103 [0.97]	-1.464 [0.04]	0.485 [0.86]	-1.923 [0.02]	0.973 [0.58]	-2.684 [0.08]	-1.451 [0.06]	-1.793 [0.08]
Observations	47,939	101,977	52,995	97,888	117,651	33,172	131,100	19,683

- Mobilization $\uparrow$ in autocracies, captured media, urban area, and areas with past conflicts

Takeaways and discussion

Mobile phones help mobilization, but primarily through channeling grievances

- When grievances do not arise, effects are insignificant
- Effects of mobile coverage *interacted with GDP (economic downturns)* significant
- Effects replicated at individual level participation outcomes
- Thus, mobile phones facilitate mobilization by channeling grievances

Discussions and remaining questions

- Mobile phones decrease cost of acquiring information and improves coordination
 - ▶ Required for mobilization and other collective actions
- Results generalizable beyond timeframe and region?
 - ▶ What about states with advanced capacities and high technological level?

Guriev, Melnikov, Zhuravskaya (2021): 3G Internet and Confidence in Government

Q: How does mobile network affect confidence in government?

Independent political thinking vs empowerment of nondemocratic regimes?

- Increase awareness and overcome collective action problems
- Easier dissemination of fake news, propaganda, and surveillance
 - ▶ Former weakens confidence in government, latter strengthens it (esp nondemocracies)

This paper analyzes how expansion of 3G networks affect confidence in government

- 840,000 individuals across 2,232 regions in 116 countries from 2008-2017
- 3G $\uparrow \rightarrow$ Support for government $\downarrow$
 - ▶ Awareness of corruption $\uparrow$, less trusting in institutions
 - ▶ Network channels outrage against status quo contagiously
- Effect dominant when internet is not censored and traditional media is censored
- Makes people aware of misgovernance: Effect present in highly corrupt countries

Data and empirical strategy

Data sources

- Government approval: Gallup World Poll (2008-2017)
 - ▶ Yes/No on confidence in national government, judiciary, elections, corruption perception
- 3G coverage: GSM association (same as Manacorda and Tesei (2020))
- Institutional quality: Global Incidence of Corruption Index, Panama Papers Database (by journalists at ICIJ), Freedom House censorship index
- Lightning: World Wide Lightning Location Network (WWLLN)

Empirical strategy leverages 3G rollout variation across time and regions

$$Y_{irt} = \gamma_1 3G_{rt} + \gamma_2 \text{Development}_{rt} + \lambda X_{irt} + \phi_r + \delta_t + \epsilon_{irt}$$

- Lightning used to instrument 3G coverage

$$3G_{rt} = \beta_1 \text{ln}_r \times t \times \text{Rich}_{cr} + \beta_2 \text{ln}_{rt} \times \text{Poor}_{cr} + \mu X_{irt} + \phi_r + \delta_t + \nu_{irt}$$

Expansion of 3G networks, 2007-2018

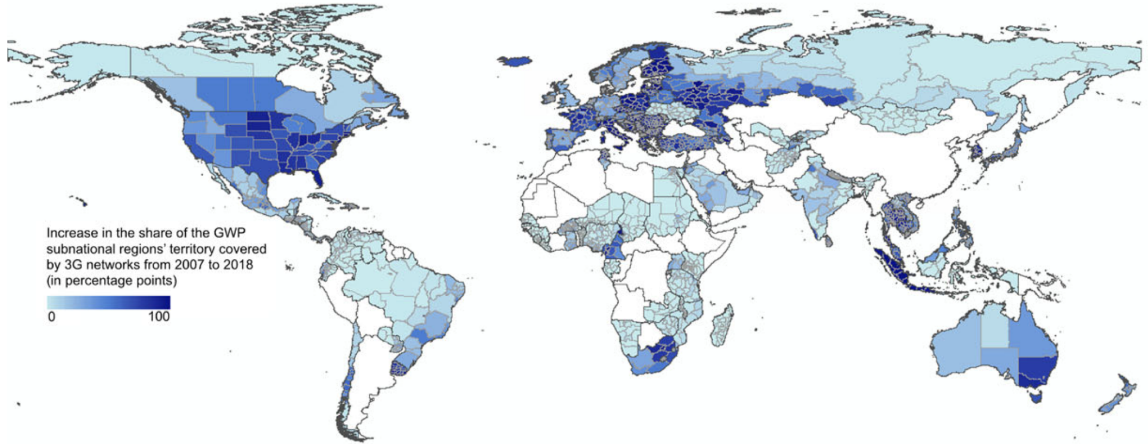


Figure: 3G network expansion in 2007-2018

Confidence in governance decreases across the board

Dep. var.:	Confidence in national government (1)	Confidence in judicial system (2)	Honesty of elections (3)	No corruption in government (4)	Share of questions with positive responses (5)
Panel A: Sample of all respondents					
Regional 3G coverage	-0.063*** (0.021)	-0.040*** (0.015)	-0.079*** (0.021)	-0.036** (0.014)	-0.056*** (0.015)
Observations	772,353	748,471	732,856	722,768	617,863
R-squared	0.164	0.163	0.168	0.225	0.242
Mean dep. var	0.439	0.534	0.505	0.226	0.432
Mean 3G coverage	0.397	0.381	0.383	0.383	0.381
Number of countries	111	116	112	112	110
Panel B: Subsample of rural residents					
Regional 3G coverage	-0.091*** (0.024)	-0.058*** (0.017)	-0.115*** (0.026)	-0.054*** (0.016)	-0.080*** (0.018)
Observations	464,831	448,449	440,786	432,460	371,055
R-squared	0.171	0.157	0.161	0.194	0.224
Mean dep. var	0.349	0.556	0.516	0.215	0.445
Mean 3G coverage	0.329	0.314	0.316	0.316	0.311
Number of countries	110	115	111	111	109
Subnational region and year FEs	✓	✓	✓	✓	✓
Baseline controls	✓	✓	✓	✓	✓

More so with uncensored internets and censored traditional media

Dep. var.:	Confidence in national government (1)	Confidence in judicial system (2)	Honesty of elections (3)	No corruption in government (4)	Share of questions with positive responses (5)
Panel C: Time-variant dummies for internet censorship and above-median press censorship					
Regional 3G coverage	-0.032 (0.029)	-0.027 (0.020)	-0.089*** (0.024)	-0.026 (0.021)	-0.046** (0.020)
Regional 3G coverage × × Internet censorship dummy	0.157*** (0.044)	0.059** (0.030)	0.195*** (0.046)	0.078** (0.031)	0.121*** (0.035)
Regional 3G coverage × × Above-median press censorship dummy	-0.116*** (0.034)	-0.051** (0.023)	-0.046 (0.030)	-0.049* (0.025)	-0.059** (0.025)
Internet censorship dummy	0.057* (0.032)	0.037 (0.024)	0.049 (0.031)	0.005 (0.023)	0.039 (0.027)
Above-median press censorship dummy	0.123*** (0.034)	0.023 (0.021)	0.070** (0.030)	0.059** (0.025)	0.068*** (0.023)
Observations	656,015	631,606	618,480	613,737	521,632
R-squared	0.158	0.166	0.158	0.234	0.239

Decrease in approval driven by corruption detection

Dep. var:	Perception of no corruption in government					
	All countries		Excluding low-income countries			
	(1)	(2)	(3)	(4)	(5)	(6)
Countries in the sample:						
Regional 3G coverage ×						
× number of Panama Papers entities per capita	−0.035**					
	(0.014)					
× number of Panama Papers entities per capita × before Panama Papers		−0.031**	−0.033**			
		(0.014)	(0.014)			
× number of Panama Papers entities per capita × after Panama Papers		−0.037**	−0.048***			
		(0.018)	(0.017)			
× 1[Top 10% of countries by Panama Papers entities per capita] × before Panama Papers				−0.045		
				(0.033)		
× 1[Top 10% of countries by Panama Papers entities per capita] × after Panama Papers				−0.100**		
				(0.040)		
× number of Panama Papers entities × before Panama Papers					−0.012***	
					(0.004)	
× number of Panama Papers entities × after Panama Papers					−0.017***	
					(0.005)	
× 1[Top 10% of countries by Panama Papers entities] × before Panama Papers						−0.092***
						(0.028)
× 1[Top 10% of countries by Panama Papers entities] × after Panama Papers						−0.174***
						(0.038)

Key takeaways and discussions

Major findings in this paper

- 3G expansion decreases confidence in government
- Effects depends on censorship and corruption levels
 - ▶ Internet censorship raises confidence, traditional media censorship decreases confidence
 - ▶ Effect stronger in more corrupt countries (by detecting new episodes of corruption)
- Electoral consequences: Voters disillusioned with incumbents
 - ▶ Vote share for incumbent ↓ Anti-establishment populists ↑

Discussions and remaining questions

- Who benefits from the 3G network?
 - ▶ Nonpopulist oppositions did not gain vote shares
 - ▶ Anti-establishment includes extreme left and right-wing parties
- What explains why anti-establishment are well off?
 - ▶ What contents (potentially misinformation)?, Effects and mechanisms of misinformation?

Enikolopov, Marakin, Petrova (2020):
Social Media and Protest Participation:
Evidence from Russia

Q: Does social media alleviate collective action problems?

Overcoming collective action problems depends on ability to communicate

- Social media allows conversations without intermediaries at low cost
- This may lead to enhanced dissemination of information
- However, if social media panders to rulers, information may decrease protests
- Challenges: Selection bias in social media usage (individual, geographic characteristics)

This paper leverages unexpected protests and features of social media (VK) in Russia

- Leverages political protests in Russia in 2011
- Protests more likely in areas where those with close ties to first users of VK lived
- Distinguish information vs collective action channels
 - ▶ Information: Increase awareness to those who do not know
 - ▶ Collective action: Facilitate coordination of logistics and peer pressure

How social media facilitates protests and application to Russia

Two channels explaining effects of social media on protests

- Overall effect: Providing precise information + Assist in coordinating logistics
 - ▶ Information: Negative content → Protest ↑, Positive content → Protest ↓
 - ▶ Coordination: Always raises protests
- Coordination augmented by city size: Overcomes large coordination costs in cities
- If critical mass required: Protests occur only if certain threshold of users is reached

Russian protests in 2011 and Vkontakte (VK)

- Internet usage uncensored in 2011 (increased censoring after protests since 2012)
- VK: First users personally approved by founder Pavel Durov - primarily SPSU students
 - ▶ Fun fact: Same person who found Telegram
- Protests in 2011 against parliamentary election fraud in Dec 2011
 - ▶ VK users more aware of election fraud and coordinated protests

It is the election where this happened...



Data sources and empirical strategy

Data sources (covering 625 cities in Russia)

- City of residence of VK users who joined before 2011 summer
- Hand collected protest info (newspapers and online resources, 12/2012 - 05/2012)
 - ▶ Complemented with alternative source from police and organizers
- VK founder cohort matched based on birth year, university, year of study (public)
- Election data from Central Election Commission

Empirical strategy uses exogenous variation based on location of VK founder's cohort

$$\text{OLS} : Y_i = \beta_0 + \beta_1 VK_i + \beta_2 X_i + e_i$$

$$\text{IV} : VK_i = \gamma_0 + \gamma_1 f(\text{same SPSU cohort as VK founder})_i + \gamma_2 X_i + u_i$$

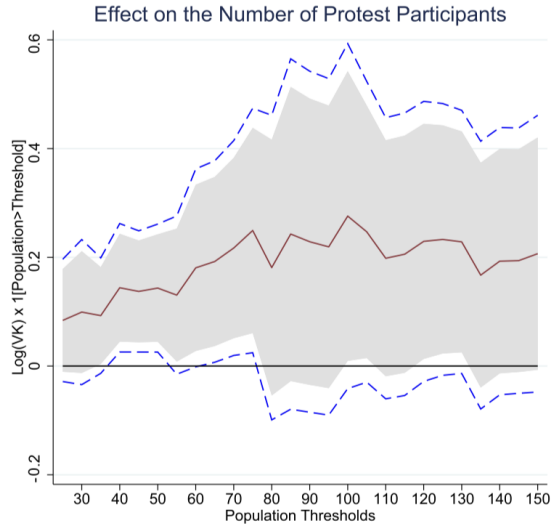
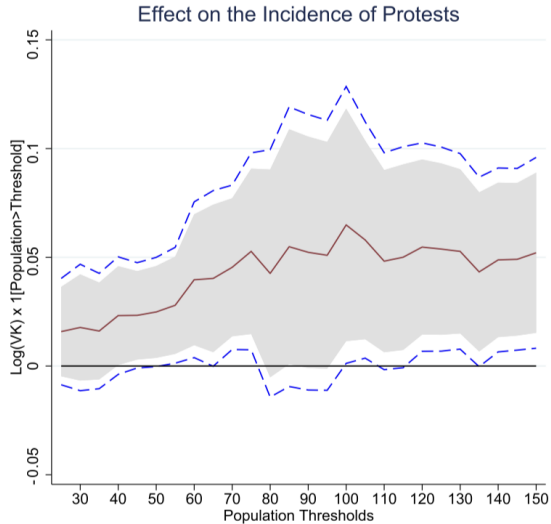
- Verifies that same cohort as VK founder determines (not earlier or later) VK usage

VK penetration, instrumented with VK founder cohort, raises protests

Panel A. Probability of Protests

	Incidence of Protests, Dummy, Dec 2011							
	IV (1)	IV (2)	IV (3)	IV (4)	OLS (5)	OLS (6)	OLS (7)	OLS (8)
Log (number of VK users), June 2011	0.466 [0.189]	0.451 [0.177]	0.458 [0.175]	0.479 [0.181]	0.060 [0.018]	0.057 [0.018]	0.055 [0.019]	0.065 [0.018]
<i>Weak IV Robust 95% Confidence Interval</i>	<i>(0.18; 1.77) (0.18; 1.56) (0.18; 1.42) (0.20; 1.53)</i>							
Log (SPbSU students), one cohort younger than VK founder	0.027 [0.024]	0.026 [0.024]	0.028 [0.025]	0.030 [0.025]	0.029 [0.021]	0.028 [0.020]	0.026 [0.021]	0.030 [0.020]
Log (SPbSU students), one cohort older than VK founder	-0.033 [0.031]	-0.029 [0.029]	-0.028 [0.027]	-0.026 [0.029]	0.003 [0.018]	0.005 [0.017]	0.003 [0.017]	0.007 [0.018]
Observations	625	625	625	625	625	625	625	625
Mean of the dependent variable	0.134	0.134	0.134	0.134	0.134	0.134	0.134	0.134
SD of the dependent variable	0.341	0.341	0.341	0.341	0.341	0.341	0.341	0.341
Population controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age cohort controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Electoral controls, 1995		Yes				Yes		
Electoral controls, 1999			Yes				Yes	
Electoral controls, 2003				Yes				Yes
Kleibergen-Paap F-statistics	6.554	6.779	7.591	7.031				
Effective F-statistics (Montiel Olea and Pflueger (2013))	10.97	12.03	12.30	12.17				

VK penetration effects larger in bigger cities



Takeaways and remaining questions

Other notable results

- Electoral outcomes: VK penetration *raised vote share* of ruling party
 - ▶ On average, more positive contents describing ruling party documented
 - ▶ Voting outcomes only affected by information effect: You vote alone
 - ▶ Protests depends on information and coordination effects
 - ▶ Thus, in this case, information and coordination effects worked in opposite directions

Takeaways and remaining questions

- Social media does increase protests, primarily through coordination channel
 - ▶ Information effect could still repress protests if government propaganda dominates
- If different contents exist in social media, which one dominates?
 - ▶ Could be a key to understanding misinformation propagation
 - ▶ Also necessary key to understand polarization on internet?

Gagliarducci, Onorato, Sobbrio, Tabellini (2020):
War of the Waves: Radio and Resistance During
World War II [student presentation]

Takeaways and remaining questions

There are many ways to overcome collective action problems

- Role of leaders to stimulate a cause and organize a movements
- Media's potential to bring individuals for a common cause
 - ▶ Bring new and accurate information to those who were unaware of it previously
 - ▶ Lower the cost of coordination: Make it easy to reach out to people
 - ▶ Internet can achieve this if it is uncensored and traditional media is censored

Remaining questions

- Leadership can be used to motivate bad things (Tianyi Wang 2021 AER)
 - ▶ Effective ways to hold decision-makers accountable is the key
- Likewise, media could channel negative influence
 - ▶ Misinformation, echo-chamber effect (or both)
 - ▶ What are the ways to mitigate or control misinformation?